

Publications (Sorted YW 2025-2017 with corresponding IF-2025 & Abstract)

1. Aditya Kurdekar, Sai Manohar Chelli, Mohan Kumar Haleyrur Giri Setty et al. Engineering Europium Nanoparticle-HIV-1 p24 conjugates as a Novel Platform Rapid and Sensitive Antibody Detection. (2025) *Talanta Open*. 11, 100451 (IF: 4.2)

Abstract: The detection and quantification of immune markers in biological samples play a critical role in various fields, including diagnostics, biomedical research, and therapeutics. Western blotting or enzyme-linked immunosorbent assays (ELISAs), which are labour-intensive and call for complicated processes, are frequently used in traditional immunodetection techniques. However, fluorescent nanoparticle-antigen conjugates have become an innovative method for immunogenic detection thanks to the recent developments in nanotechnology. This study demonstrates the fabrication and working principle of conjugates of fluorescent nanoparticles and HIV-1 p24 in antibody detection. Europium nanoparticles are the fluorophore conjugated to the HIV-1 p24 antigen, a capsid protein. Computational studies were conducted to determine whether the bioconjugation would be feasible, and the findings showed high stability of the conjugates. After validation, the bioconjugation of HIV-1 p24 and the nanoparticle was carried out using the EDC-NHS approach. Studies employing anti-HIV-1 p24 antibodies demonstrated that Europium nanoparticle-p24 (EuNP-p24) conjugates could achieve a sub-picogram-level sensitivity of 0.4 pg/mL and a very high recovery rate of $\geq 92\%$, indicating their potential for improving the sensitivity of the assay and its efficiency in clinical diagnostics.

2. Vemula, Sai Krishna; Chelli, Sai Manohar; Biswas, Chinmoy et al. Adsorption and electron injection studies in N719 sensitized Ag⁻, Au⁻ implanted and O₂ annealed titania films. (2024). *Optical Materials*. 154, 115633 (IF: 3.8)

Abstract: The device performance of novel solar cell configurations with porphyrin, hybrid perovskite-based materials as sensitizers depends on the interfacial processes and engineered interfaces with characteristic material properties can be fabricated employing energetic ion implantations. In this experimental work, the dye adsorption, optical absorption and interfacial electron injection in N719 sensitized Ag⁻, Au⁻, sequentially impinged and O₂ annealed titania films were studied. Prior to the dye sensitization process, impinged samples were fabricated employing mono-ion type, sequential implantations and a set of the as-implanted films were annealed at 450 °C for 1 h in O₂ environment. The N719 molecules are adsorbed onto Ag⁻, Au⁻ impinged films through SCN ligands, indicated by the appearance of new Raman modes $\nu(\text{Ag-S})$, $\nu(\text{Au-S})$ respectively. The ¹MLCT, ³MLCT bands of the dye molecules are altered upon implantation and annealing processes. A fast interfacial electron injection process with $\tau_2 \sim 3.7$ ps has been observed in the sensitized annealed sequentially impinged film at 6×10^{15} ions cm⁻² (sensitized pristine TiO₂ exhibits $\tau_2 \sim 51$ ps) and is attributed to Ag⁻, Au⁻ implants introducing new states to the conduction band of anatase TiO₂. The ability of ion-implantation technique to introduce new states to the band structure of materials has enormous significance in photo-catalytic, water-splitting, solar cell and hot electron injection technologies.

3. Aditya Dileep Kurdekar, Prajal Chettri, Rajasimha Kurnoothala, Chelli Sai Manohar et al. Investigation of enhanced third-order optical nonlinearity in novel coenzyme A capped silver nanoparticles. (2023) *Journal of Materials Chemistry, C*. 11, 16234-16246 (IF: 5.7)

Abstract: In this paper, we report the design and synthesis of novel coenzyme A (CoA)-capped silver nanoparticles and the investigation of their third-order nonlinear optical properties. Computational studies were carried out to understand the feasibility of forming CoA-capped nanoparticles, which indicated successful stabilization of silver nanoparticles upon surface functionalization with coenzyme A. The synthesis of CoA-capped silver nanoparticles was carried out by chemical reduction of silver nitrate by sodium borohydride in the presence of heat. The

appearance of the surface plasmon resonance peak at 429 nm confirmed the formation of silver nanoparticles. TEM, SEM, DLS, and Zeta potential analyses showed the formation of polydispered spherical-shaped nanoparticles with CoA capping. The third-order nonlinear susceptibilities were measured using the degenerate four-wave mixing (DFWM) technique. The CoA-capped silver nanoparticles exhibited a higher nonlinear susceptibility than the traditionally used borohydride-capped silver nanoparticles. The enhanced nonlinearity was attributed to the localised surface plasmon resonance (LSPR) of the silver nanoparticles and the electron density distribution at the interface of the nanoparticles and the capping agent. The optical properties of the CoA-capped nanoparticles can open up avenues for the use of metal nanoparticles with bulkier capping agents in optoelectronic applications.

4. Alagumalai, Ananthan; Venu Rajendran, Muthukumar; Ganesan, Saraswathi; Sudhakaran Menon, Vidya; Raman, Rohith Kumar; Chelli, Sai Manohar et al. Interface Modification with Holistically Designed Push–Pull D– π –A Organic Small Molecule Facilitates Band Alignment Engineering, Efficient Defect Passivation, and Enhanced Hydrophobicity in Mixed Cation Planar Perovskite Solar Cells. (2022) *ACS Applied Energy Materials*. 5:6, 6783–6796 (IF: 5.5)

Abstract: In perovskite solar cells, interfaces play a significant role in determining the device stability and device performance. Here, we introduce a versatile donor– π –acceptor (D– π –A) based organic small molecule (AA1) containing phenothiazine (PTZ) with a long alkyl chain as the donor unit, the vinyl-substituted thiophene moiety as a π bridge, and a rhodanine-(CN)₂ moiety as an acceptor unit for the first time, and it was successfully deployed to passivate the defects at the surface and grain boundaries of a dual-cation perovskite absorber. The synthesized organic small molecule was characterized thoroughly using ¹H NMR, ¹³C NMR, FT-IR, UV–vis, CV, TGA, and HRMS studies. The FT-IR spectral analysis and X-ray photoelectron spectroscopy (XPS) analysis confirm the interaction between the organic small molecule and the perovskite absorber. Simulated electrostatic potential surface (EPS) images obtained through the density functional theory (DFT) study reveal higher electron density over the acceptor unit of AA1, which ensures effective perovskite defect passivation. The formation of high-quality perovskite film with enhanced crystallinity, improved grain size, and band energy level alignment leads to effective charge carrier transport. The dual nature (defect passivation and optimized band energy level alignment) of AA1 passivation increases the surface photovoltage (from ~100 to ~155 mV) and reduces the defect density and ideality factor ($\sim 1.94 \times 10^{14} \text{ cm}^{-3}$ from $\sim 7.1 \times 10^{14} \text{ cm}^{-3}$ and ~ 1.64 from ~ 1.88 , respectively). The nonradiative recombination is suppressed along with reduced hysteresis, which leads to higher open-circuit voltage (1.09 V from 1.05 V) and power conversion efficiency. This work highlights the use of a push–pull small organic molecule which ensures effective passivation of undercoordinated Pb²⁺ defects and improved power conversion efficiency. The superior hydrophobic nature of the molecule results in device robustness. Finally, this all-in-one molecule wrestles the three major challenges commonly seen in perovskite solar cells, i.e., interface improvement, unhindered charge carrier transport, and device stability.

5. Pawar, Gaurav; Mahammad Ghouse, Shaik; Kar, Swayamsiddha; Manohar Chelli, Sai et al. SmI₂-mediated C-alkylation of Ketones with Alcohols under Microwave Conditions: A Novel Route to Alkylated Ketones. (2022) *Chemistry–An Asian Journal*. 17:8, e202200041 (IF: 3.5)

Abstract: A novel protocol is developed towards the preparation of alkylated ketones from alcohols in presence of catalytic amount of SmI₂ and base with the elimination of water as a single by-product under microwave irradiation conditions. Furthermore, applicability of this methodology to the synthesis of Donepezil and late-stage functionalization in Pregnenolone is also reported. Successful application of this methodology in Friedländer quinolone synthesis using 2-aminobenzyl alcohol and various acetophenones expand the synthetic utility of this protocol.

6. Kar, Swayamsiddha; Rai, Prashant; Chelli, Sai Manohar; Akhir, Abdul et al. Exploring the unexpected formation of spirobibenzopyrans and benzopyrylium salts and effect of Lewis acids on the Claisen-Schmidt reaction. (2021) *Journal of Molecular Structure*. 1240, 130598 (IF: 4.0)

Abstract: Unexpected spiro cyclic products were obtained from the popular alkali catalyzed Claisen-Schmidt reaction of ketones with salicylaldehyde apart from the bis-chalcones. However, in the acid catalyzed reaction, we observe their transformation to a benzopyrylium salt. In the present study, we demonstrate this phenomenon for three ketones namely cyclopentanone, cyclohexanone and cycloheptanone. Each of the corresponding products obtained were characterized using UV-Vis, FT-IR, ^1H NMR, ^{13}C NMR, and mass spectrometry. In addition, the structure of the spiro molecule synthesized from cycloheptanone reactant has been determined by X-ray crystallography and further investigated for its thermal properties using TGA/DTA technique. To rationalize this phenomenon, we first studied the effect of solvents on the reaction, and further explored the effect of Lewis acid catalyst and various solvents for the same reaction conditions. Further, we have evaluated *in silico*, the impact of thermodynamic parameter like entropy. In this regard, we explored the relationship between the bis-chalcone, spiro molecule and the benzopyrylium salt. Consequently, the effect of the ring expansion from cyclopentanone to cycloheptanone was investigated. In conclusion, we have presented the anti-bacterial potential of the synthesized spirobibenzopyrans and the benzopyrylium salts.

7. Malasala, Satyaveni; Polomoni, Anusha; Chelli, Sai Manohar et al. A microwave-assisted copper-mediated tandem approach for fused quinazoline derivatives. (2021) *Organic & Biomolecular Chemistry*. 19:8,1854-1859 (IF: 2.9)

Abstract: A method for the microwave-assisted copper-mediated oxidative coupling reaction of different aldehydes and quinazolines/benzimidazoles has been developed for the synthesis of fused-polycyclic systems *via* new C–N bond formation. The current methodology involves the use of environmentally benign NH_4OAc as the amine source in the presence of 2-propanol as the solvent. This novel tandem reaction approach offers a rapid and straightforward access to complex fused quinazoline derivatives in an efficient manner.

8. Bhandari, Rajni; Nehra, Anita; Manohar, Chelli Sai et al. Zirconium–Cerium and Zirconium–Lanthanum complexed polyvinyl alcohol films for efficient fluoride removal from aqueous solution. (2021) *Journal of Dispersion Science and Technology*. 42:10, 1550-1565 (IF: 1.9)

Abstract: In the current work, user-friendly, biodegradable, Zirconium-Lanthanum (ZLPC) and Zirconium-Cerium (ZCPC) complexed Polyvinyl alcohol (PVA) films were solvent casted for effective defluoridation of water. The films were characterised using SEM coupled EDS, XPS, TGA, XRD and FT-IR. These films adsorbed 95–97% fluoride from water at pH 7, with 85% removal by ZLPC and 93% removal by ZCPC within the first 45 minutes of the adsorption process. A maximum adsorption capacity of 11.57 mg/g and 12.88 mg/g, corresponding to ZLPC and ZCPC respectively, was obtained using Langmuir adsorption isotherm. The films followed pseudo-second-order kinetics ($R^2 \leq 0.99$) in both the cases, which was further established by low values of chi-square and root mean square error (RMSE). The ICP-AES analysis confirmed no leaching of the metal ions from the polymer films into the treated water. The films were found to be stable in both acidic and basic medium. No interference to fluoride adsorption was detected even at 100 mg/L of Cl^- , NO_3^- and SO_4^{2-} while some interference was observed in the presence of PO_4^{3-} and HCO_3^- . The change in Gibbs free energy for fluoride adsorption using the ab-initio method was found to be -330.05 kJ/mol, which indicated spontaneity of the process. This was further reinforced by the thermodynamic data obtained from adsorption studies. The films were reusable and showed ~85% removal of fluoride even after the fifth regeneration cycle. Thus, these handy films have a scope to be developed into a prototype for defluoridation of ground water.

9. Kurdekar, Aditya Dileep; Manohar, Chelli Sai; Chunduri, LA Avinash et al. Computational design and clinical demonstration of a copper nanocluster based universal immunosensor for sensitive diagnostics. (2020) *Nanoscale Advances*. 2:1, 304-314 (IF: 4.6)

Abstract: Nanoparticle based sensors are good alternatives for non-enzymatic sensing applications due to their high stability, superior photoluminescence, biocompatibility and ease of fabrication, with the only disadvantage being the cost of the synthesis process (owing to the expensive precursors and infrastructure). For the first time, we report the design of an immunosensor employing streptavidin conjugated copper nanocluster, developed at a much lower cost compared to other nanomaterials like noble metal nanoparticles and quantum dots. Using *in silico* tools, we have tried to establish the dynamics of conjugation of nanocluster to the streptavidin protein, based on EDC-NHS coupling. The computational simulations have successfully explained the crucial role played by the components of the immunosensor leading to an efficient design capable of high sensitivity. In order to demonstrate the functioning of the Copper Nanocluster ImmunoSensor (CuNIS), HIV-1 p24 biomarker test was chosen as the model assay. The immunosensor was able to achieve an analytical limit of detection of 23.8 pg mL^{-1} for HIV-1 p24 with a linear dynamic range of $27\text{--}1000 \text{ pg mL}^{-1}$. When tested with clinical plasma samples, CuNIS based p24 assay showed 100% specificity towards HIV-1 p24. With the capability of multiplexed detection and a cost of fabrication 100 times lower than that of the conventional metal nanoclusters, CuNIS has the potential to be an essential low-cost diagnostic tool in resource-limited settings.

10. Manohar, Chelli Sai; Kumar, B Siva; Sadhu, Sai Pavan Prashanth; Srimadh, Sai Krishna et al. Novel Lead-free biocompatible piezoelectric Hydroxyapatite (HA)–BCZT ($\text{Ba}_{0.85}\text{Ca}_{0.15}\text{Zr}_{0.1}\text{Ti}_{0.9}\text{O}_3$) nanocrystal composites for bone regeneration. (2019) *Nanotechnology Reviews*. 8:1, 61-78 (IF: 6.1)

Abstract: BCZT ($\text{Ba}_{0.85}\text{Ca}_{0.15}\text{Zr}_{0.1}\text{Ti}_{0.9}\text{O}_3$) is a recent class of lead-free ferroelectric material associated with high piezoelectric coefficient, making it suitable to inspire hydroxyapatite (HA)-BCZT ceramics for bone materials. Nano-crystalline hydroxyapatite (HA) synthesized using the hydrothermal route was characterized via FT-IR, Raman spectroscopy, X-ray powder diffraction (XRD), and Scanning Electron Microscopy (SEM). We also rationalized its formation as a function of operating conditions such as dwell time and temperature in this route. The nano-crystalline BCZT powder was synthesized via a sol-gel technique and its structural and morphological characterization were carried out using Raman Spectroscopy, XRD and Transmission Electron Microscopy (TEM). These investigations facilitated the optimization of HA-BCZT compositions and their electrical poling conditions to achieve enhanced piezoelectric effect. The composites (HA-BCZT) sintered at 1350°C exhibited promising piezoelectric properties. We report the enhanced piezoelectric coefficient (d_{33}) of $7 \pm 1 \text{ pC/N}$ for 50% HA-BCZT which is significant as compared to that reported in the literature for $\sim 98\%$ BT (barium titanate) -HA composites. We highlight the role of Simulated Body Fluid (SBF) on the intriguing phase-change of Tricalcium Phosphate (TCP) obtained at this sintering temperature, to hydroxyapatite for its essential contribution to promote bone growth. We theoretically support the confirmed *in vitro* biocompatibility of these composites.

11. Hariharan, Abishek; Chelli, Sai Manohar; Belliraj, Siva Kumar et al. Microfluidics based SERS substrate for PPB level detection of catechol. (2019) *Optical Materials*. 94: 305-310 (IF: 3.8)

Abstract: The application of microfluidics for largescale rapid analytics holds great promise in the pharmaceutical diagnostics and analytical chemistry. Here we report a paper based microfluidic substrate designed by the impregnation of Silver nanoparticles. This study demonstrates the achievement of a thousand-fold increase in the successful detection level up to 10 ppb (90.8 nM) by employing Surface Enhanced Raman Spectroscopy (SERS) for the detection of Catechol. The presented sensor exhibits the following main features: (i) high specificity of enzyme (Hemocyanin)-based sensing, (ii) effective SERS sensitivity, (iii) easiness and cost-effectiveness of a paper-based

platform. We rationalize these findings using the *ab initio* DFT simulations using Gaussian 09 whose theoretical calculations reflect the observed experimental Raman peaks.

12. Chelli, Sai Manohar; Gupta, Parth; Belliraj, Siva Kumar. An in silico design of bioavailability for kinase inhibitors evaluating the mechanistic rationale in the CYP metabolism of erlotinib. (2019) ***Journal of Molecular Modelling***. 25,1-1 (IF: 1.9)

Abstract: Soft spot analysis helps evaluate the site of the metabolic lability that impacts the bio-availability of the drug. However, given its laborious and time consuming experimentation, we propose a reliable and cheap in silico strategy. In this context, we hypothesized a mechanistic rationale for metabolism of erlotinib by the CYP3A4 enzyme. The comparison of the 3D conformations of the target CYP class of enzymes using MD simulations with GROMACS helped evaluate its impact on the metabolism. The molecular docking studies using Autodock-Vina ascertained the explicit role of the Fe ion present in the Heme moiety in this process. This mechanism was confirmed with respect to 13 other popular approved FDA kinase inhibitors using *ab initio* DFT calculations using Gaussian 09 (G09), molecular docking studies with Autodock-Vina, and MD simulations with GROMACS. We then developed a quantitative (Q-Met) metabolic profile of these soft spots in the molecules and demonstrated the lack of a linear relationship between the extent of metabolism and drug efficacy. We thus propose an economic in silico strategy for the early prediction of the lability in kinase inhibitors to help model their bio-availability and activity simultaneously, prior to clinical testing.

13. Gupta, Agman; Manohar, Chelli Sai; Siva Kumar, Belliraj. Optimization of cation-doped nano-hydroxyapatite in combination with amorphous aluminum hydroxide for defluoridation. (2019) ***Water Supply***. 19:6, 1686-1694 (IF: 1.9)

Abstract: Given widespread fluoride in the ground water, there is a need for effective defluoridation in several geographical areas. In this regard, we explored heavily doped cationic nano-composites of hydroxyapatite (HA) given its surface chemistry for adsorption of the specific anion. We synthesized and extensively characterized HA nano-rods (HA-NR), Al/Mg-HA nanocomposites and amorphous aluminum hydroxide, and optimized their efficient defluoridation. The kinetics and thermodynamics of adsorption were further evaluated to establish the mechanistic rationale and its spontaneity. We report the optimized ideal adsorbents for the near-total removal of fluoride that demonstrated 99.99% and 99.98% efficiency with adsorption capacities of 83.3 and 81.3mg/g respectively. The adsorbent composites were (Mg-HA)-Al(OH)₃ and (HA-NR)-Al(OH)₃ in 1:1 ratio. The optimal conditions for defluoridation were 25mg of adsorbent in 25ml (10mg/L) fluoride solution at room temperature agitated for 10h in the pH range of 4.88–7.20.

14. Kurdekar, Aditya Dileep; Avinash Chunduri, LA; Manohar, C Sai et al. Streptavidin-conjugated gold nanoclusters as ultrasensitive fluorescent sensors for early diagnosis of HIV infection. (2018) ***Science Advances***. 4:11, eaar6280 (IF: 11.7)

Abstract: We have engineered streptavidin-labeled fluorescent gold nanoclusters to develop a gold nanocluster immunoassay (GNCIA) for the early and sensitive detection of HIV infection. We performed computational simulations on the mechanism of interaction between the nanoclusters and the streptavidin protein via in silico studies and showed that gold nanoclusters enhance the binding to the protein, by enhancing interaction between the Au atoms and the specific active site residues, compared to other metal nanoclusters. We also evaluated the role of glutathione conjugation in binding to gold nanoclusters with streptavidin. As proof of concept, GNCIA achieved a sensitivity limit of detection of HIV-1 p24 antigen in clinical specimens of 5 pg/ml, with a detection range up to 1000 pg/ml in a linear dose-dependent manner. GNCIA demonstrated a threefold higher sensitivity and specificity compared to enzyme-linked immunosorbent assay for the detection of HIV p24 antigen. The specificity of the immunoassay was 100% when tested with

plasma samples negative for HIV-1 p24 antigen and positive for viruses such as hepatitis B virus, hepatitis C virus, and dengue. GNCIA could be developed into a universal labeling technology using the relevant capture and detector antibodies for the specific detection of antigens of various pathogens in the future.

15. Manohar, Chelli Sai; Manikandan, A; Sridhar, P et al. Drug repurposing of novel quinoline acetohydrazide derivatives as potent COX-2 inhibitors and anti-cancer agents. (2018) ***Journal of Molecular Structure***. 1154, 437-44 (IF: 4.0)

Abstract: Novel QuinolineAcetohydrazide (QAh) derivatives (9a-n) were firstly evaluated *in silico* to determine their anti-inflammatory and anti-cancer efficacy via the mechanisms of COX1 and COX2 inhibition, and NF- κ B, HDAC and Human Topoisomerase I pathways respectively. In the studied set, the trifluoro substituted QAh derivatives: (E)-N'-(4-(trifluoro methyl) benzylidene)-2-(7-fluoro-2-methoxy quinolin-8-yl) acetohydrazid and (E)-N'-(3-(trifluoro methyl) benzylidene)-2-(7-fluoro-2-methoxy quinolin-8-yl) acetohydrazide are determined to be potential leads, indicated from their best docked scores, relative ligand efficiency, and significant structural attributes evaluated by *ab initio* simulations. The only setback being their partition co-efficient that retrieved a red flag in the evaluation of their Lipinski parameters. The experimental *in vitro* studies confirmed the significant enhancement as COX-2 inhibitors and appreciable enhancement in MTT assay of breast and skin cancer cell lines. Significantly, trifluoro substituent in the quinoline scaffold can be reasoned to note the excellent binding affinity to all the evaluated drug targets.

16. Kurdekar, Aditya Dileep; Chunduri, LA Avinash; Chelli, Sai Manohar et al. Fluorescent silver nanoparticle based highly sensitive immunoassay for early detection of HIV infection (2017) ***RSC advances***. 7: 32, 19863-19877 (IF: 3.9)

Abstract: For the first time, we have engineered streptavidin labeled fluorescent silver nanoparticles for their application in immunosensing of biomolecules which will significantly increase sensitivity without compromising the specificity. A computational perspective for understanding the efficiency of the conjugation process is provided as a proof of concept. The fluorescence based sandwich immunoassay was demonstrated to detect HIV-1 p24 antigen in clinical specimens with improved sensitivity and specificity. The detection range of the fluorescent silver nanoparticle-based immunoassay (FSNIA) was found to be between 10 and 1000 pg mL⁻¹ in a linear dose dependent manner. False positives were not observed with plasma samples from healthy adults (HIV^{-ve}), hepatitis B (HBV^{+ve}) and hepatitis C (HCV^{+ve}). Plasma samples that were HIV^{-ve} showed no interference with detection of HIV-1 p24 antigen. This technology can be used in resource limited settings and easily adopted for the detection of other pathogen antigens.

17. Badiya, Pradeep Kumar; Patnaik, Sai Gourang; Srinivasan, Venkatesh; Reddy, Narendra; Manohar, Chelli Sai et al. Ag-protein plasmonic architectures for surface plasmon-coupled emission enhancements and Fabry-Perot mode-coupled directional fluorescence emission. (2017) ***Chemical Physics Letters***. 685, 139-145 (IF: 2.8)

Abstract: We report the use of silver decorated plant proteins as spacer material for augmented surface plasmon-coupled emission (120-fold enhancement) and plasmon-enhanced Raman scattering. We extracted several proteins from different plant sources [*Triticum aestivum* (TA), *Aegle marmelos* (AM), *Ricinus communis* (RC), *Jatropha curcas* (JC) and *Simarouba glauca* (SG)] followed by evaluation of their optical properties and simulations to rationalize observed surface plasmon resonance. Since the properties exhibited by protein thin films is currently gaining research interest, we have also carried out simulation studies with Ag-protein biocomposites as spacer materials in metal-dielectric-metal planar microcavity architecture for guided emission of Fabry-Perot mode-coupled fluorescence.

18. Sridhar, G; Mohammed Bilal, I; Easwaramoorthy, D; Kutti Rani, S; Siva Kumar, B; Manohar, Chelli Sai. Synthesis, characterization and antimicrobial activities of copper, nickel, cobalt, chromium complexes derived from (Z)-4-Fluoro-N-(2, 7-dimethylhept-6-enylidene) benzenamine. (2017) *Journal of the Brazilian Chemical Society*. 28:5, 756-767 (IF: 1.3)

Abstract: Melonal is a relatively unexplored ingredient in the field of chemistry with excellent sensory properties. The synthesis and characterization of novel transition 3d metal complexes of copper(II), nickel(II), cobalt(II) and chromium(III) involving the Schiff base of melonal were explored to examine their biological activity. Characterization of the complexes was carried out using matrix-assisted laser desorption/ionization-time-of-flight (MALDI-TOF), gas chromatography-mass spectrometry (GC-MS), thermogravimetric analysis (TGA), UV-Visible and infrared (IR) spectrophotometry. The antimicrobial studies were conducted against six bacterial strains and six fungi. The minimum inhibition concentration observed was compared against the standard antibiotic gentamycin and the anti-fungal drug amphotericin. The activity studies indicated that cobalt(II) complex exhibited activity better than standard drug amphotericin against *Penicillium chrysogenum*. Molecular docking study confirms the protein binding and supports the experimental finding. Binuclear cobalt(II) and chromium(III) bridging complexes of Schiff base ligand were obtained.